

hi, I'm

ANANYA Penumudi

B.Tech CSE (IoT) · Embedded systems, IoT & systems engineering

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📍 Hyderabad, India



✳ About

B.Tech (CSE-IoT) student working across embedded systems, AI-powered devices and systems-engineering research. I trained at BHEL Bengaluru and wrote a thesis on KAVACH, the Automatic Train Protection system on Indian Railways. I like building cyber-physical systems that sense, decide and act.

✳ Experience

Industrial Trainee

Bharat Heavy Electricals Limited (BHEL), Bengaluru

Jun – Jul 2026

- Trained at the Solar Business Division: manufacturing, production workflows, QA and system integration in a Government of India PSU.
- Studied railway safety technology, signalling infrastructure and KAVACH through technical discussions and documentation.

Internship certificate ➔

Software Engineering Intern · Position Tracker

Alfago

Jan 2026

- Implemented tax-lot based PnL calculations and backend transaction logic for a tracker of transaction- and portfolio-level profit and loss.

Internship certificate ➔

Operations Intern (remote)

Kartgen Infotech LLP

Jan – Apr 2025

- Supported business development and operations: process coordination and stakeholder communication, under Shrivatsa Malviya.

Letter of recommendation ➔

✳ Featured work

Elementium AI: virtual chemistry lab

Team project · React Three Fiber · FastAPI · MongoDB Atlas · Gemini

2026

- 360° 3D lab with drag-and-drop apparatus, real-time titration and colour-change chemistry.
- Gemini-powered tutor plus a rule-based engine that catches unsafe or out-of-order steps; auto-filled lab notebook.

GitHub ➔

Beyond Collision Prevention: a systems-engineering analysis of KAVACH

Research thesis · industry guidance from BHEL Bengaluru

Jun 2026

- Analysed the architecture, operating modes, communication framework and functional-safety mechanisms of India's indigenous ATP system.
- Built an interactive simulator of KAVACH's modes and a digital twin.

Thesis report ➔ · Mode simulator ➔ · Digital twin ➔

✳ Education

2023 – now

VNR Vignana Jyothi Institute of Engineering & Technology

B.Tech, CSE (IoT) · JNTUH

CGPA 8.92 / 10 (till Sem VI)

2021 – 2023

Sri Chaitanya Junior College

Intermediate · Telangana State Board

97.4%

2021

Bharatiya Vidya Bhavan's Public School

High School · CBSE

GPA 9.38 / 10

✳ Skills

IOT & EMBEDDED

Arduino ESP32 SPI I2C
PWM BLE GPS GSM
Accelerometers

PROGRAMMING

C C++ Python Java
JavaScript R HTML/CSS
DSA

AI & DATA

Computer vision
Edge inference
Mood detection
Anomaly detection Tableau

SYSTEMS ENGINEERING

Requirements analysis
Functional safety V&V
System architecture
Traceability

* More projects

Smart AI Wheelchair System

Jan 2026

IoT + edge AI

- Vision-based obstacle detection with automatic braking for collision prevention.
- GPS geofencing, caregiver alerts and sensor fusion across camera, IMU and proximity sensors.

[Technical report ↗](#)

AI-Enhanced Secure Data Wiping Platform

Nov 2025

AI + cybersecurity · team project

Data sanitisation using binary overwrite, with cryptographic verification of every wipe and AI-based anomaly detection.

[GitHub ↗](#)

Syncope: fall detection & alert system

Aug 2025

Wearable IoT

Wearable that detects falls from motion-sensor data and alerts caregivers automatically.

[Documentation ↗](#)

Auralyn: emotional-wellness wearable

2025 – now

IoT + AI + healthcare · in design

Designing an ESP32-S3 wearable with BLE that pairs biometric sensing with mood analysis, with apps for the wearer, their therapist and family.

FocusFlow: ADHD-friendly productivity platform

Apr 2025

AI + web

Mood tracking, journaling, facial-expression analysis and gamified focus tools.

[GitHub ↗](#)

Ideavoid

Nov 2024

Web platform

Startup-idea validation with peer feedback and market-viability assessment.

Cheugy

Mar 2024

Wellness app

Stress management and self-care activities.

[GitHub ↗](#)

Syndicate

Dec 2023

Film-based learning platform

Gamified learning through films, documentaries, quizzes and discussions.

* Achievements

2nd Prize, IDEATHON 2023

VNRVJIET · [Certificate ↗](#)

Smart India Hackathon 2025

Solution design, prototyping and technical documentation

24-hour Hackathon, Convergence 2025

VNRVJIET · [Certificate ↗](#)

24-hour Hackathon, DEMUX 2024

BVRIT · [Certificate ↗](#)

National Conclave 2025

CBIT · [Certificate ↗](#)

* Certifications

DevOps with Real-Time Practical Exposure

Brainovision · AICTE approved · Jul – Nov 2025

Tableau Certified Data Analyst Training

Udemy · 59 hours · Jun 2024 · [Certificate ↗](#)

RoboJAM Robotics Workshop

Convergence 2K23, VNRVJIET · Dec 2023

* Activities

Computer Society of India

Member, VNRVJIET chapter · 2023 – now. Organises workshops, coding events and inter-college competitions.

Art of Living Cultural Club

Member · 2023 – now. College events and volunteering.

* Tools & cloud

AWS MySQL MongoDB
CI/CD Figma AutoCAD
StarUML MS Excel

* Languages

English Hindi Telugu